

Gabrielle Johnson's Feasibility Memo

<https://github.com/gabriellejohnson784/Gabrielle-Johnson-Carisurg-Portfolio.git>

The dataset is **unfeasible** for training an AI assisted triage model at Mercer General ED.

Dataset Summary

The dataset contains 55,121 emergency department patient encounters across 225 columns, approximately 200 of which are chief complaint columns. The remaining columns were grouped by clinical vitals, demographic features, administrative features, and the ESI levels. Potential leakage variables, such as disposition, were kept separate because they occur after triage and should not be used for prediction.

The dataset is adult only, as none of the patients are under 18 years old, with approximately 32% being over the age of 65. In terms of demographics, women are the majority of the patients recorded at 58%, additionally, 80% of patients are non-hispanic with 53% being of the White/Caucasian race.

Triage acuity was recorded using the Emergency Severity Index (ESI), which ranges from 1 to 5, where 1 represents the most urgent cases and 5 represents the least urgent cases. Nearly half (49%) of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1.

The dataset also includes routinely collected triage vital signs, including heart rate, systolic blood pressure, diastolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose. Vital signs were reviewed using both plausible clinical limits and normal reference ranges. 29 patients had clinically implausible vital signs which were flagged as possible data quality issues.

The most common chief complaint was abdominal pain, representing 12.8% of encounters, followed by "other" at 7.8%. Although no demographic variables, vital signs, or ESI levels were missing, 52 patients had no recorded chief complaint. These cases were reviewed separately because the absence of a chief complaint may reflect documentation gaps rather than true absence of symptoms.

Top 3 Quality Concerns

1. Missingness and incomplete safety variables

Some important triage and safety variables are missing or unavailable. Pain score is not present, even though it is clinically useful for triage decision support at Mercer. Pregnancy status is also unavailable, which limits the ability to test the proposed Mercer safety guardrails for pregnant patients. Although no demographic variables, vital signs, or ESI levels were missing in the cleaned dataset, 52 patients had no recorded chief complaint, which may reflect incomplete documentation rather than true absence of symptoms. In addition, 29 patients had vital signs outside clinically plausible ranges. Although these cases represented less than 1% of the dataset, they remain important because incomplete chief complaint data and extreme outliers may distort model training

Mitigation: Patients with no chief complaint and clinically implausible vital signs are flagged for separate review rather than automatically removed, so that further analysis can be done to determine whether they represent documentation errors or clinically meaningful edge cases

2. Triage Level Imbalance

The dataset is heavily concentrated around mid acuity patients, with nearly half of all encounters classified as ESI 3 and fewer than 1% classified as ESI 1. This creates a concern because the model may learn the common ESI categories better than the rare but most urgent cases. ESI specific performance, especially for ESI 1, should be tested separately rather than relying on overall accuracy.

Mitigation: Training methods should consider class weighting for ESI 1

3. Representativeness and transferability risk

The dataset is adult only and the racial distribution is majority White/Caucasian. This may not reflect Mercer General ED's Caribbean patient population, where demographic and chronic illness patterns will differ. Because of this, the dataset should only be treated as exploratory and should not be used without local validation.

Mitigation: Treat the model as an exploratory prototype and test performance on local data before deployment.

Top 3 Reasons Not to Proceed

1. Poor transferability to Mercer

The dataset is adult only, with no patients under 18, and the racial distribution is majority White/Caucasian. This limits transferability as Mercer serves a Caribbean population with a contrasting racial profile, mixed age population and experiences different environmental, and disease pattern contexts. Tropical conditions, local chronic disease, and race/ethnicity may affect chief complaint patterns, acuity, and triage decisions.

2. Important proposed triage variables are missing and safety guardrails

Pain score is not available in the dataset, even though it would be clinically useful for triage decision support. Pregnancy status is also unavailable, which limits the ability to build the safety checks proposed for pregnant patients. This is important because the proposed Mercer system includes additional safeguards for paediatric, elderly and pregnant patient groups.

3. Severe class imbalance may weaken ESI 1 prediction and vague/missing chief complaint category may weaken the model's clinical understanding

Nearly half of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1. This creates a class imbalance concern because a model may achieve high overall accuracy by predicting common ESI categories while performing poorly on rare but clinically critical ESI 1 cases. Since ESI 1 patients represent the most urgent cases it's crucially important for the model to be able to detect high acuity patients.

The most common chief complaint was abdominal pain at 12.8%, followed by "other" at 7.8%. Because "other" is the second largest chief complaint category, this may hide clinically decisive symptoms inside one vague label. This may weaken model training because many different symptoms are grouped under "other," making it harder for the model to learn clear chief complaint patterns.

A small number of patients also had no recorded chief complaint or clinically implausible vital signs. Although these cases represent less than 1% of the dataset, they still matter because incomplete or extreme records can distort model training.

Caveats

The dataset may still be useful for exploratory analysis, including reviewing vital sign patterns, chief complaint relationships, and ESI distributions, but it is not sufficient for deployment without local validation. Paediatric safety and Caribbean population generalizability remain unresolved because the dataset is adult only and majority White/Caucasian. Pain score and pregnancy status are missing, which limits the ability to test the proposed Mercer safety guardrails. The “other” chief complaint category is the second most common complaint, reducing clinical specificity and potentially weakening the model’s ability to learn clear chief complaint patterns.

ESI 1 cases make up less than 1% of encounters, so the model may fail to identify the most urgent cases. Simple Spearman correlation also showed weak individual associations between vital signs and ESI, suggesting that no single vital sign strongly explains triage level on its own. However, vital signs may still contribute predictive value when combined with chief complaints and demographics.

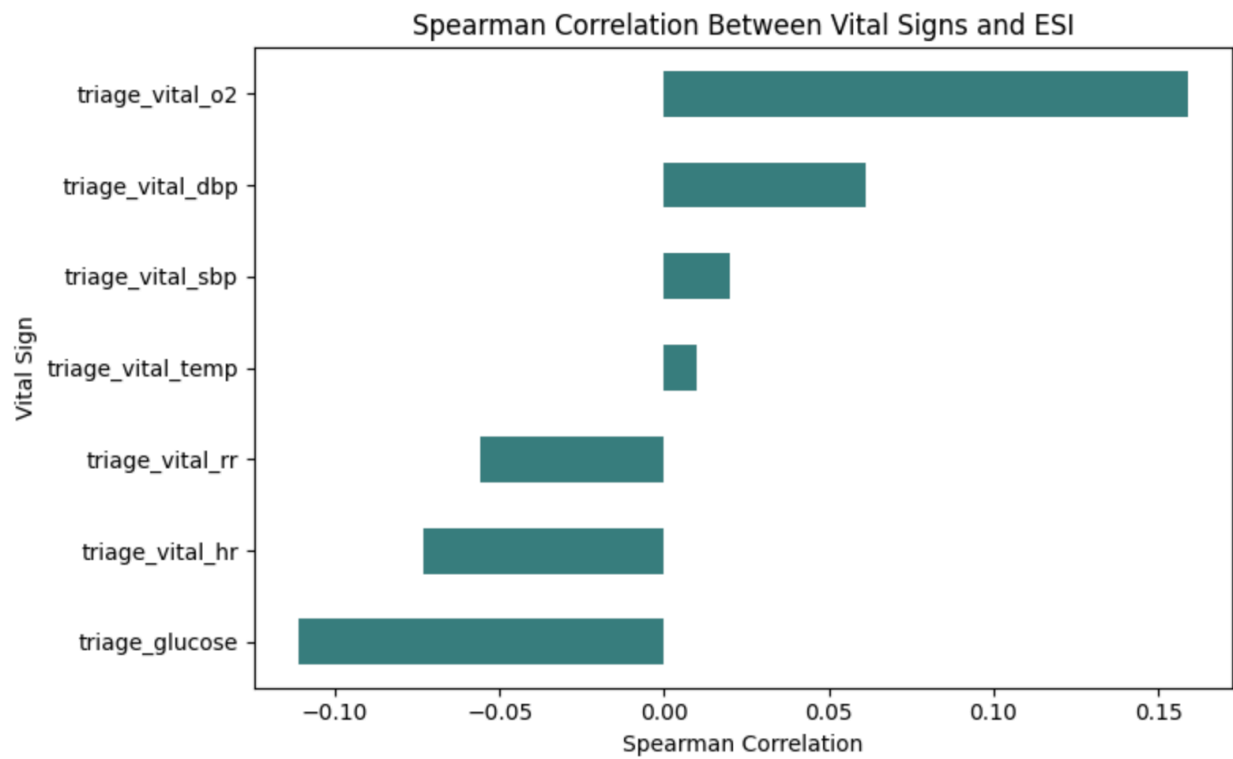
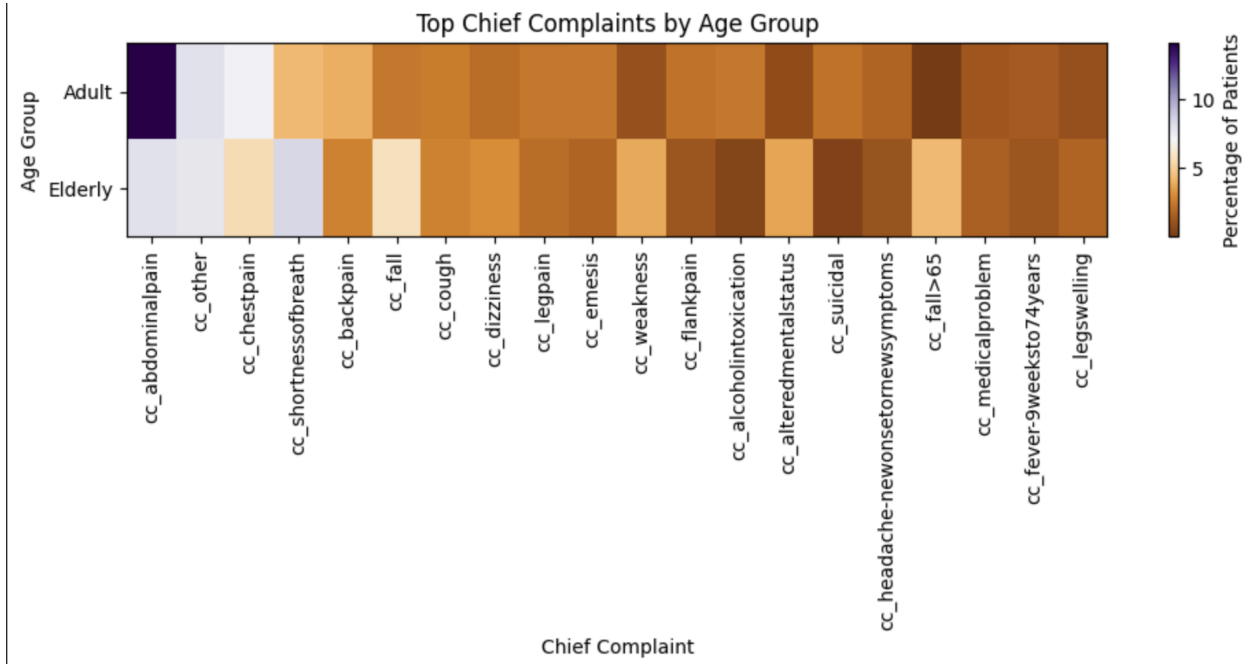
Chief complaints may help explain ESI when considered individually, but the dataset does not show strong evidence that combinations of complaints or the total number of complaints reliably predict acuity.

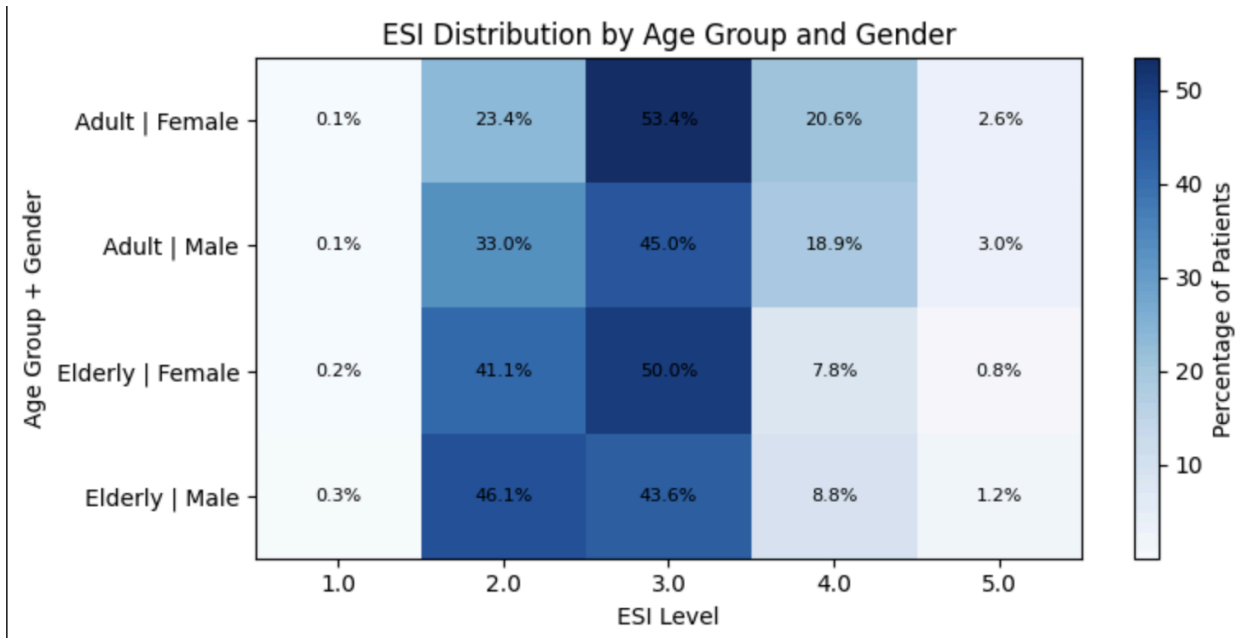
Patients with no recorded chief complaint flags and patients with clinically implausible vital signs should also be reviewed separately as they could hold clinical meaning, because small numbers of incomplete or extreme records can still affect model reliability.

Top 10 Feature list

Feature	Justification
Arrival Mode	Arrival mode was selected because it is one of the earliest variables available in the triage workflow and often reflects pre-triage urgency, with the arrival-mode heatmap showing much higher ESI 2 proportions for ambulance, wheelchair, and police arrivals than for car or walk-in arrivals.
Chief Complaint	Chief complaint was selected because it captures the patient's presenting problem early in triage and helps guide initial acuity classification, with the ESI distribution plot showing that different complaints clustered differently across ESI levels.(1)
Age	older patients/pediatrics are more likely to deteriorate or present atypically, and the data exploration showed age had one of the stronger observed relationships with ESI, with elderly groups having higher proportions of ESI 2 encounters.
triage_vital_o2	Oxygen saturation was selected because it directly reflects respiratory problems, and it had the strongest Spearman correlation with ESI among the vital signs at approximately 0.159.(1)
triage_vital_rr	Respiratory rate was selected because abnormal breathing rate can signal

	respiratory distress, sepsis, shock, or instability, making it clinically important even though its Spearman correlation with ESI was weak. (1)
triage_vital_hr	Heart rate was selected because tachycardia or bradycardia can result in life threatening complications(1)
triage_vital_sbp	Systolic blood pressure was selected because very low or very high values can indicate cardiovascular instability, or severe physiological stress, even though its simple correlation with ESI was weak in this dataset.(1)
triage_glucose	Glucose was selected because severe hypo- or hyperglycaemia can seizures, or indicate diabetic emergencies, and it had the second strongest vital-sign Spearman correlation with ESI at approximately -0.111 .
triage_temp	Temperature was selected because abnormal body temperature can indicate infection or inflammations, although its simple Spearman correlation with ESI was weak in this dataset, it is best interpreted alongside other vital signs and chief complaints.(1)
Gender	Gender was selected because the exploration showed visible ESI distribution differences by age group and gender, especially among elderly males.





References:

1. Emergency_Severity_Index_Handbook.pdf [Internet]. [cited 2026 Jul 7]. Available from: https://media.emscimprovement.center/documents/Emergency_Severity_Index_Handbook.pdf?utm_source=chatgpt.com